Summary and context

This dataset contains 77 student records with 11 fields capturing basic demographics, prior education, study behavior and assessment results. Key columns are: first/last name, Age, gender, country, residence, entryEXAM score, prevEducation, studyHOURS, and assessment scores in Python and DB. The goal of the analysis was to understand data quality, distributions of scores and study hours, and relationships between entry exam and course outcomes.

Headline findings

- The dataset is small (77 records) and largely complete — only the Python score has 2 missing values.
- Most students are from Norway and many live in on-campus or private housing, but categorical values need cleaning (mixed gender labels and residence variants).
- Study hours cluster tightly around high values (roughly 140–160 hours). Entry exam and assessment scores skew toward higher values.
- Entry exam shows a moderate positive relationship with both Python and DB scores; Python and DB are positively associated with each other.

Data snapshot

Below are the first five rows so you can see column names and example records:

fNAME	lNAME	Age	gender	country	residence	entryE
Christina	Binger	44	Female	Norway	Private	72

fNAME	lNAME	Age	gender	country	residence	entryE
Alex	Walekhwa	60	M	Kenya	Private	79
Philip	Leo	25	Male	Uganda	Sognsvann	55
Shoni	Hlongwane	22	F	Rsa	Sognsvann	40
Maria	Kedibone	23	Female	South Africa	Sognsvann	65

(From the original notebook head output.)

Data size and completeness

- Records: 77 rows, 11 columns.
- Missing values: Python has 2 missing entries; other columns are complete.
- Gender entries include multiple variants (Female, Male, M, F, female) indicating inconsistent labeling that should be standardized.
- Residence labels also have variants (e.g., "BI Residence", "BI-Residence", "BIResidence") that will benefit from normalization.

Supporting table: Missingness counts (column order preserved)

Column	Missing count
fNAME	0
lNAME	0

Column	Missing count
Age	0
gender	0
country	0
residence	0
entryEXAM	0
prevEducation	0
studyHOURS	0
Python	2
DB	0

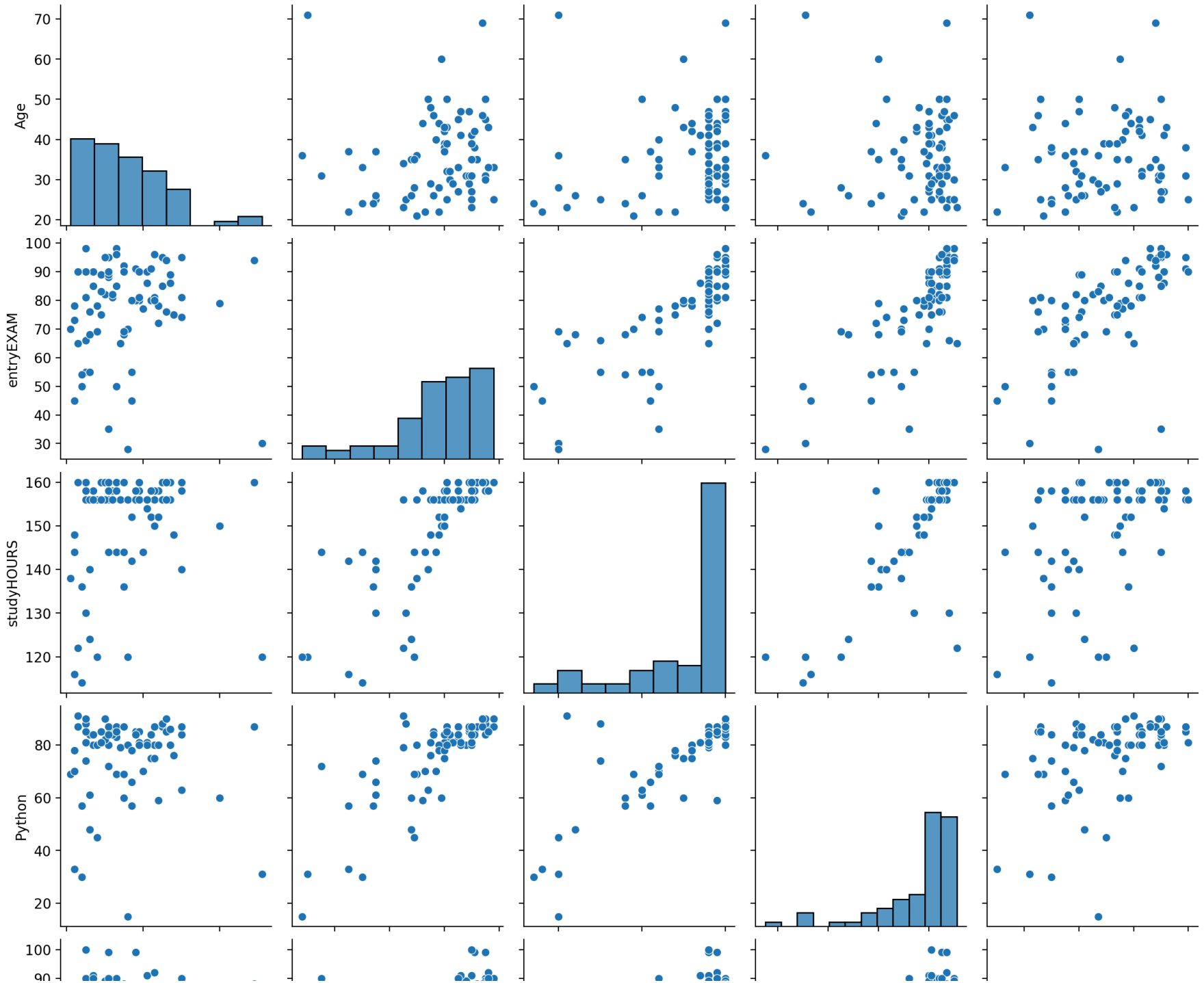
Key distributions (what the charts show)

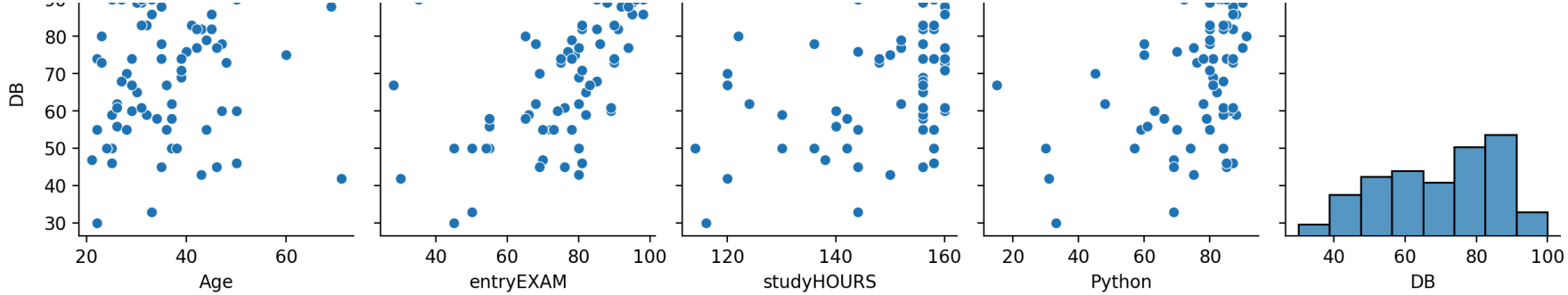
- Age: wide range (min 21, max 71), mean ~35, SD ~10 — a mixed-age cohort.
- Entry Exam: mean ~76.8, median 80, distribution skewed toward higher scores (min 28, max 98).
- Study Hours: concentrated high values (mean ~149.7, IQR narrow: 144–158), indicating most students reported substantial study time.
- Python and DB scores: Python mean ~75.85 (some missing), DB mean ~69.47; both skewed toward upper ranges with maxima near 100.

Reference visualizations:

- Pairwise relationships and outliers:

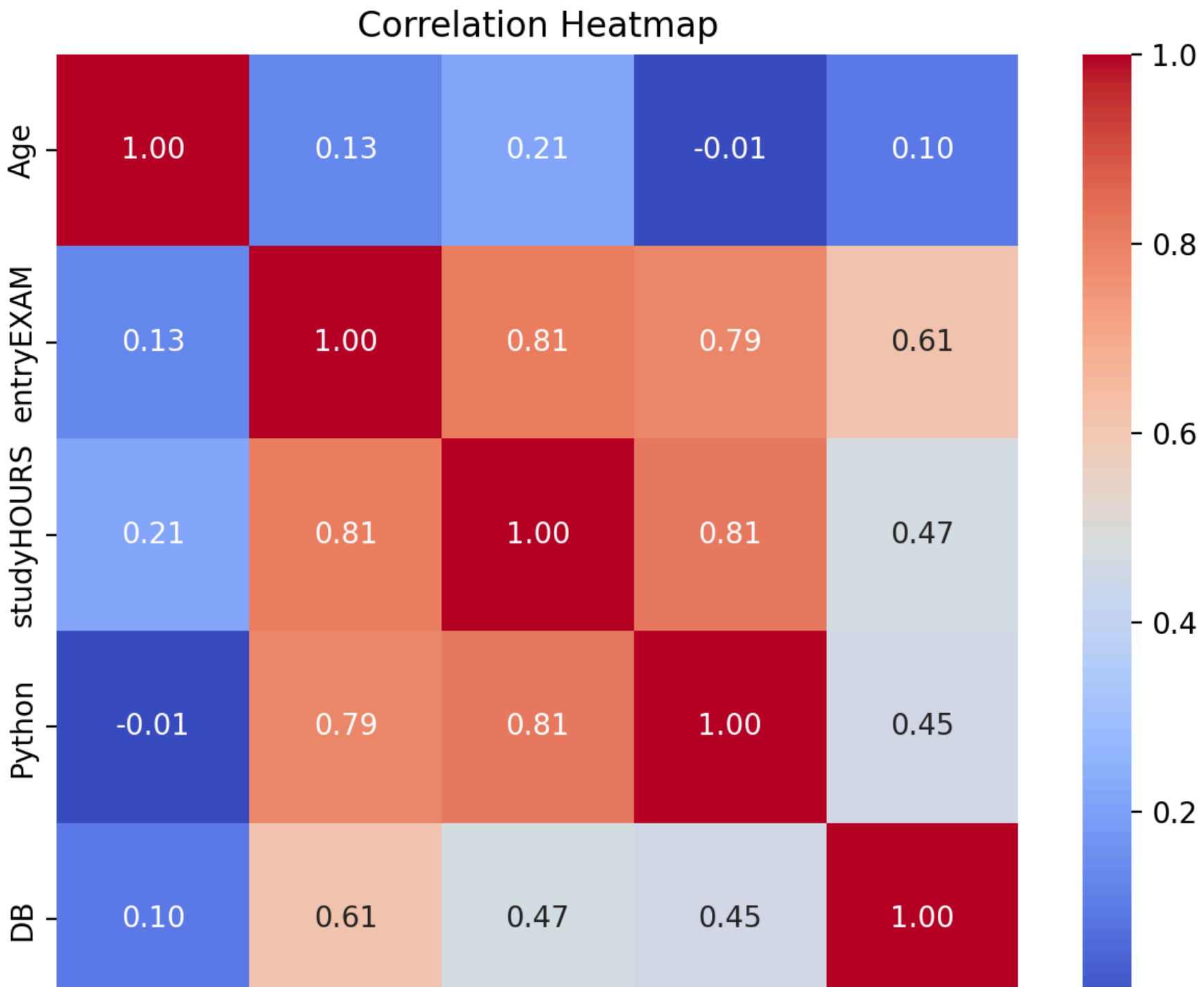
Download


Age
entryEXAM
studyHOURS
Python



- Correlation heatmap among numeric variables:

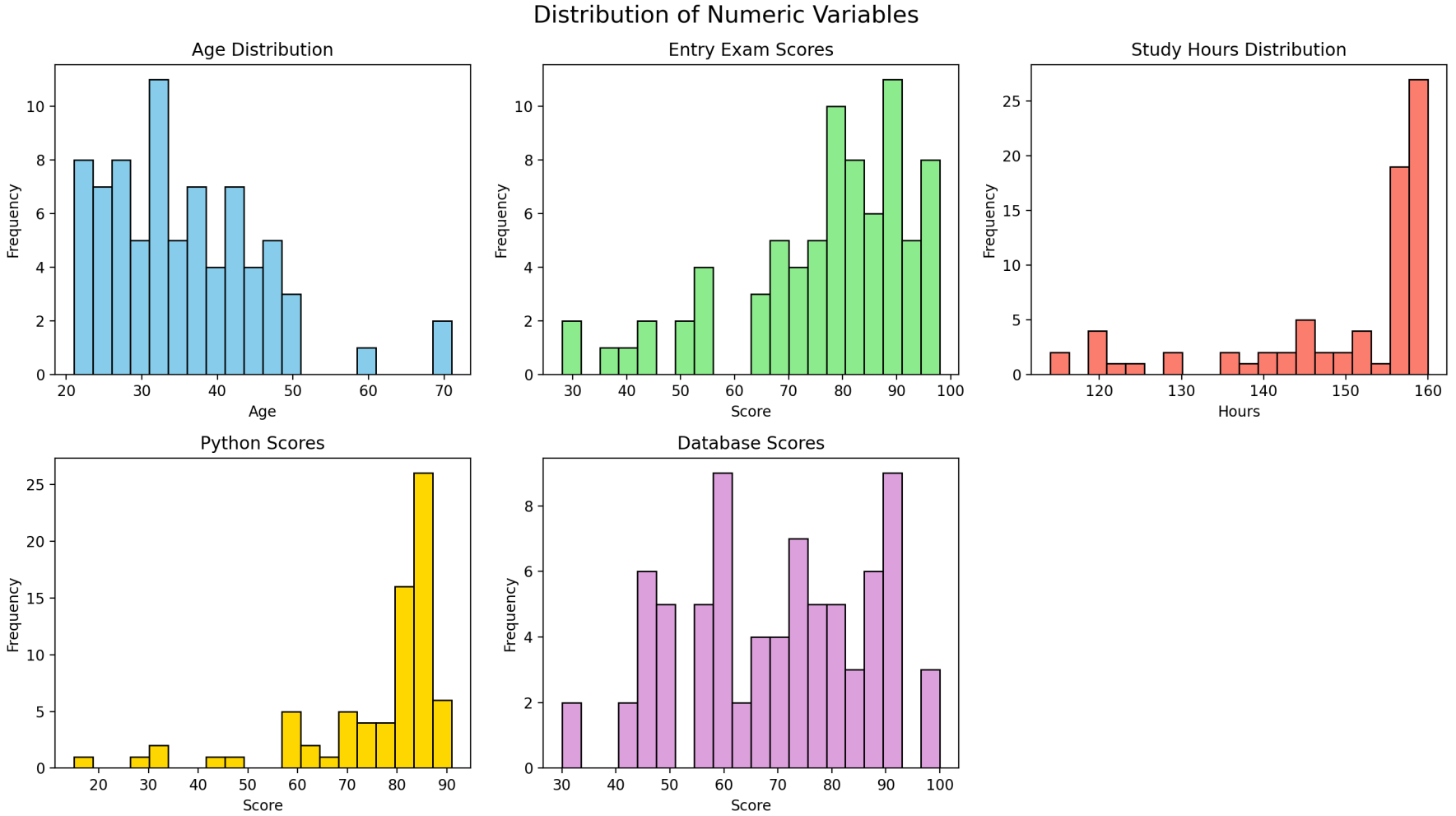
Download


Correlation Heatmap
Age
entryEXAM
studyHOURS
Python
DB
1.00
0.13
0.21
-0.01
0.10
0.13
1.00
0.81
0.79
0.61
0.21
0.81
1.00
0.81
0.47
-0.01
0.79
0.81
1.00
0.45
0.10
0.61
0.47
0.45
1.00
1.0
0.8
0.6
0.4
0.2



- Individual distributions for Age, entryEXAM, studyHOURS, Python, DB:

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Notable patterns and supporting data

- Entry exam correlates positively with course outcomes:
  - The heatmap and pairplot reveal a moderate positive relationship between entryEXAM and both Python and DB scores. This suggests students with stronger entry exam results tend to perform better in assessments.
  - Numeric summary: entryEXAM mean 76.75, Python mean 75.85, DB mean 69.47 — all clustered at relatively high values.
- Python and DB performance move together:
  - Pairwise plots show a positive association between Python and DB scores, indicating students who do well in one assessment often do well in the other.
- Study hours are high and narrowly distributed:
  - studyHOURS mean ~149.7 with a tight IQR (144–158). Most students report similarly high study effort, so differences in outcomes are less likely to be explained by study hours alone.
- Categorical imbalances and cleaning needs:
  - Gender counts show majority labeled as Female (including variants); Male labels appear in several forms. Example top counts:

gender	count
Female	41

gender	count
Male	32
M	1
F	1
female	1

- Country distribution is heavily skewed to Norway (47 of 77), with a few students from other countries. Residence values also require normalization:

country	count
Norway	47
Uganda	4
Kenya	3
Germany	3
Denmark	2

residence	count
Private	33
BI Residence	29

residence	count
Sognsvann	12
BI-Residence	1
BIResidence	1

- Prior education mix:
    - Many students have Bachelors (24), High School (18) and Masters (16), plus some Diplomas and Doctorates. This diversity could influence performance comparisons.

prevEducation	count
Bachelors	24
High School	18
Masters	16
Diploma	9
Doctorate	5

Short recommendations (based on findings)

- Standardize categorical fields (gender, residence variants, prevEducation) before any subgroup analysis to avoid misleading splits.

- Impute or review the two missing Python scores (small number; simple imputation or removal is reasonable depending on downstream needs).
- For deeper insights, run group comparisons (average Python/DB by prevEducation, residence, and country) and regression to quantify the effect of entryEXAM and studyHOURS on outcomes.
- Consider visual summaries (boxplots) of Python/DB by prevEducation and residence to surface group differences.

Conclusion

This cohort is small but informative: most students study a similar (high) number of hours, entry exam performance aligns with course outcomes, and Python and DB scores are positively linked. Cleaning categorical inconsistencies and handling two missing Python scores will sharpen further subgroup and predictive analyses. Overall, the data supports the expectation that incoming competence (entryEXAM) is a useful indicator of course performance, while study hours appear uniformly high across students.